

Catálogo do SDK do iRacing (corrida real em Road Atlanta, 60 Hz)

O catálogo lista **349 variáveis de telemetria** (47 são arrays por carro/roda) e **293 chaves de SessionInfo** em 8 seções. Fonte: `sdk_catalog_race.json`. O Live Coach V3 lê **54** variáveis pelo nome; as outras **295** estão sem uso hoje (a contagem é por nome literal no código, então é aproximada).

Guia de exemplos: o que cada informação do SDK significa

Valores tirados de `sdk_catalog_race.json` (corrida em Road Atlanta, 60 Hz). Os exemplos de bitfield foram decodificados; as descrições vêm do próprio catálogo do SDK (campo `desc`).

Como ler um array por carro

Variáveis `CarIdx*` têm 72 posições, uma por carro. O índice é o `CarIdx` do piloto na sessão (não o número do carro, e o do jogador muda entre sessões). - `CarIdxLapDistPct` = `[0.0358, 0.991, 0.0019, 0.9841, ...]`: o carro 0 está a 3,6% da volta; o 1 está a 99,1%, logo atrás da linha; o 2 acabou de cruzar. - `CarIdxEstTime` = `[1.87, 67.97, 0.11, 67.56, ...]`: segundos que o carro levaria (na volta de referência da classe) para chegar onde está. Com isso dá para estimar o gap entre dois carros mesmo sem histórico de voltas.

Bitfields decodificados

- `SessionFlags` = 268697636 → **green | blue | servicable | startHidden**. É o estado global (bandeira verde, aviso de largada etc.).
- `CarIdxSessionFlags` = 262144 = `0x40000` → só "servicable": o carro está normal. Se aparecer `0x80000` (furled) é aviso de Slow Down; `0x10000/0x20000/0x100000` (black, disqualify, repair) é ida obrigatória ao box. É o que alimenta as badges do Relative.
- `PitSvFlags` = 16 = `0x10` → o box tem "abastecer" marcado. Por isso o Fuel at End soma o `PitSvFuel`.
- `EngineWarnings` = 0 → nenhuma luz acesa (bits para água, óleo, pressão de óleo, limitador etc.).
- `CarLeftRight` = 1 → livre. 2 carro à esquerda, 3 à direita, 4 dos dois lados, 5 dois à esquerda, 6 dois à direita.
- `CarIdxTrackSurface` = 3 → na pista. -1 fora do mundo, 0 fora da pista, 1 parado no box (pit stall), 2 aproximando-se dos boxes.

Exemplos de informação para o piloto

Informação possível	Variáveis	Exemplo com o valor gravado
Delta ao vivo	<code>LapDeltaToBestLap</code> , <code>LapCurrentLapTime</code>	+1,78 s contra a melhor volta, volta atual em 7,46 s
Combustível	<code>FuelLevel</code> , <code>FuelUsePerHour</code> , <code>PitSvFuel</code>	57,5 L; consumo instantâneo 112 kg/h; 34 L planejados no box
Temperatura de pneus	<code>LFtempCL</code> (e as outras 11 de carcaça)	carcaça esquerda dianteira a 78,3 °C
Desgaste de pneu	<code>LFwearR</code>	98,6% de banda restante
Clima e pista	<code>TrackTempCrew</code> , <code>TrackWetness</code> , <code>Precipitation</code> , <code>WindDir</code>	pista a 26,7 °C, umidade nível 1, sem chuva, vento a 2,86 rad
Quem fala no rádio	<code>RadioTransmitCarIdx</code>	-1 = ninguém; se for 5, mostrar o nome do carro 5
ABS atuando	<code>BrakeABSactive</code>	false na frente reta
Incidentes	<code>PlayerCarMyIncidentCount</code>	0
Distribuição de freio	<code>dcBrakeBias</code>	47,0

Informação possível	Variáveis	Exemplo com o valor gravado
Reboque	PlayerCarTowTime	0 = sem reboque; maior que 0 = sendo rebocado

Observação: LFPressure não existe neste catálogo (a pressão de pneu só é publicada no box), então uma tela de pressão exigiria outra variável ou o SessionInfo.

Variáveis por família (usadas / total)

Família	Usadas	Total	Exemplos não usados
Outros	4	66	ManualBoost, ManualNoBoost, MemPageFaultSec, MemSoftPageFaultSec, PaceMode
CarIdx	15	27	CarIdxTrackSurface, CarIdxTrackSurfaceMaterial, CarIdxBestLapNum, CarIdxQualTireCompound, CarIdxQualTireCompoundLocked
Lap	3	27	LapCompleted, LapDist, LapBestLap, LapBestLapTime, LapCurrentLapTime
Player	4	22	PlayerCarPosition, PlayerCarClassPosition, PlayerCarClass, PlayerTrackSurfaceMaterial, PlayerCarTeamIncidentCount
Steering	1	15	SteeringFFBEnabled, SteeringWheelAngleMax, SteeringWheelPctTorque, SteeringWheelPctTorqueSign, SteeringWheelPctTorqueSignStops
LF	1	15	LFTiresUsed, LFTiresAvailable, LFbrakeLinePress, LFcoldPressure, LFodometer
RF	0	15	RFTiresUsed, RFTiresAvailable, RFbrakeLinePress, RFcoldPressure, RFodometer
LR	0	15	LRTiresUsed, LRTiresAvailable, LRbrakeLinePress, LRColdPressure, LRodometer
RR	0	15	RRTiresUsed, RRTiresAvailable, RRbrakeLinePress, RRColdPressure, RRodometer
Session	6	14	SessionTick, SessionLapsRemain, SessionLapsRemainEx, SessionTimeTotal, SessionLapsTotal
Pit	2	14	Pitch, PitsOpen, PitRepairLeft, PitOptRepairLeft, PitstopActive
Replay	0	6	ReplayFrameNum, ReplayFrameNumEnd, ReplayPlaySpeed, ReplayPlaySlowMotion, ReplaySessionTime
Tire	0	6	TireSetsUsed, TireSetsAvailable, TireLF_RumblePitch, TireRF_RumblePitch, TireLR_RumblePitch
Vel	0	6	VelocityZ_ST, VelocityY_ST, VelocityX_ST, VelocityZ, VelocityY
Is	0	5	IsReplayPlaying, IsDiskLoggingEnabled, IsDiskLoggingActive, IsInGarage, IsGarageVisible
Chan	0	4	ChanAvgLatency, ChanLatency, ChanPartnerQuality, ChanClockSkew
Yaw	0	4	Yaw, YawNorth, YawRate_ST, YawRate

Família	Usadas	Total	Exemplos não usados
Cam	0	4	CamCarIdx, CamCameraNumber, CamGroupNumber, CamCameraState
Shift	0	4	ShiftPowerPct, ShiftGrindRPM, Shifter, ShiftIndicatorPct
Fuel	3	4	FuelPress
Radio	0	3	RadioTransmitCarIdx, RadioTransmitRadioldx, RadioTransmitFrequencyIdx
Brake	1	3	BrakeRaw, BrakeABSActive
Car	1	3	CarDistAhead, CarDistBehind
Roll	0	3	Roll, RollRate_ST, RollRate
Track	2	3	TrackTempCrew
Air	1	3	AirDensity, AirPressure
Engine	0	3	EngineWarnings, Engine0_RPM, Engine1_RPM
Oil	0	3	OilTemp, OilPress, OilLevel
Push	0	2	PushToTalk, PushToPass
IsOn	1	2	IsOnTrackCar
Cpu	0	2	CpuUsageFG, CpuUsageBG
Throttle	1	2	ThrottleRaw
Clutch	1	2	ClutchRaw
Wind	2	2	
Solar	0	2	SolarAltitude, SolarAzimuth
Fast	0	2	FastRepairUsed, FastRepairAvailable
Lat	0	2	LatAccel_ST, LatAccel
Lon	0	2	LongAccel_ST, LongAccel
Water	0	2	WaterTemp, WaterLevel
Dis	0	1	DisplayUnits

Família	Usadas	Total	Exemplos não usados
Driver	0	1	DriverMarker
Frame	0	1	FrameRate
Gpu	0	1	GpuUsage
ChanQuality	0	1	ChanQuality
Gear	1	1	
Speed	1	1	
Enter	0	1	EnterExitReset
Skies	0	1	Skies
Rel	1	1	
Fog	0	1	FogLevel
Precip	1	1	
Weather	1	1	
Voltage	0	1	Voltage
Manifold	0	1	ManifoldPress

Todas as variáveis

Nome	Tipo	Qtd	Unidade	Descrição	Exemplo (valor gravado na corrida)	Usada
AirDensity	float	1	kg/m^3	Density of air at start/finish line	1.1402 kg/m^3	
AirPressure	float	1	Pa	Pressure of air at start/finish line	98123.5 Pa	
AirTemp	float	1	C	Temperature of air at start/finish line	24.1286 C	sim
Brake	float	1	%	0=brake released to 1=max pedal force	0 %	sim

Nome	Tipo	Qt d	Unidade	Descrição	Exemplo (valor gravado na corrida)	Usada
BrakeABSActive	bool	1		true if abs is currently reducing brake force pressure	False	
BrakeRaw	float	1	%	Raw brake input 0=brake released to 1=max pedal force	0 %	
CamCameraNumber	int	1		Active camera number	1	
CamCameraState	bitfield	1	irsdk_CameraState	State of camera system	80 (0x50) = auto shot selection + key acceleration	
CamCarIdx	int	1		Active camera's focus car index	0	
CamGroupNumber	int	1		Active camera group number	9	
CarDistAhead	float	1	m	Distance to first car in front of player in meters	7.3524 m	
CarDistBehind	float	1	m	Distance to first car behind player in meters	11.4388 m	
CarIdxBestLapNum	int	72		Cars best lap number	carros 0-3: [-1, -1, -1, -1]	
CarIdxBestLapTime	float	72	s	Cars best lap time	carros 0-3: [-1, -1, -1, -1] s	sim
CarIdxClass	int	72		Cars class id by car index	carros 0-3: [4029, 2523, 2523, 2523]	sim
CarIdxClassPosition	int	72		Cars class position in race by car index	carros 0-3: [5, 8, 3, 13]	sim
CarIdxEstTime	float	72	s	Estimated time to reach current location on track	carros 0-3: [1.8738, 67.9659, 0.1132, 67.5569] s	sim
CarIdxF2Time	float	72	s	Race time behind leader or fastest lap time otherwise	carros 0-3: [0.004, 0.021, 0.016, 0.026] s	sim
CarIdxFastRepairsUsed	int	72		How many fast repairs each car has used	carros 0-3: [0, 0, 0, 0]	
CarIdxGear	int	72		-1=reverse 0=neutral 1..n=current gear by car index	carros 0-3: [4, 2, 2, 1]	
CarIdxLap	int	72		Laps started by car index	carros 0-3: [1, 0, 1, 0]	sim

Nome	Tipo	Qt d	Unidade	Descrição	Exemplo (valor gravado na corrida)	Usa da
CarIdxLapCompleted	int	72		Laps completed by car index	carros 0-3: [0, -1, 0, -1]	sim
CarIdxLapDistPct	float	72	%	Percentage distance around lap by car index	carros 0-3: [0.0358, 0.991, 0.0019, 0.9841] %	sim
CarIdxLastLapTime	float	72	s	Cars last lap time	carros 0-3: [-1, -1, -1, -1] s	sim
CarIdxOnPitRoad	bool	72		On pit road between the cones by car index	carros 0-3: [False, False, False, False]	sim
CarIdxP2P_Count	int	72		Push2Pass count of usage (or remaining in Race)	carros 0-3: [-1, -1, -1, -1]	sim
CarIdxP2P_Status	bool	72		Push2Pass active or not	carros 0-3: [False, False, False, False]	sim
CarIdxPaceFlags	bitfield	72	irsdk_PaceFlags	Pacing status flags for each car	carro do jogador: 0 (0x0) = nenhum bit	
CarIdxPaceLine	int	72		What line cars are pacing in or -1 if not pacing	carros 0-3: [-1, -1, -1, -1]	
CarIdxPaceRow	int	72		What row cars are pacing in or -1 if not pacing	carros 0-3: [-1, -1, -1, -1]	
CarIdxPosition	int	72		Cars position in race by car index	carros 0-3: [5, 22, 17, 27]	sim
CarIdxQualTireCompound	int	72		Cars Qual tire compound	carros 0-3: [0, 0, 0, -1]	
CarIdxQualTireCompoundLocked	bool	72		Cars Qual tire compound is locked-in	carros 0-3: [False, False, False, False]	
CarIdxRPM	float	72	revs/min	Engine rpm by car index	carros 0-3: [7621.1, 6167.2, 7153.6, 7299.94] revs/min	
CarIdxSessionFlags	bitfield	72	irsdk_Flags	Session flags for each player	carro do jogador: 262144 (0x40000) = servicable	sim
CarIdxSteer	float	72	rad	Steering wheel angle by car index	carros 0-3: [0.0639, 0.0374, 0.0239, -0.5523] rad	

Nome	Tipo	Qt d	Unidade	Descrição	Exemplo (valor gravado na corrida)	Usada
CarIdxTireCompound	int	72		Cars current tire compound	carros 0-3: [0, 0, 0, 0]	sim
CarIdxTrackSurface	int	72	irsdk_TrkLoc	Track surface type by car index	carro do jogador: 3 = OnTrack	
CarIdxTrackSurfaceMaterial	int	72	irsdk_TrkSurf	Track surface material type by car index	carros 0-3: [1, 1, 1, 1]	
CarLeftRight	int	1	irsdk_CarLeftRight	Notify if car is to the left or right of driver	1 = Clear	sim
ChanAvgLatency	float	1	s	Communications average latency	0.0333 s	
ChanClockSkew	float	1	s	Communications server clock skew	0 s	
ChanLatency	float	1	s	Communications latency	0.0333 s	
ChanPartnerQuality	float	1	%	Partner communications quality	1 %	
ChanQuality	float	1	%	Communications quality	1.0001 %	
Clutch	float	1	%	0=disengaged to 1=fully engaged	1 %	sim
ClutchRaw	float	1	%	Raw clutch input 0=disengaged to 1=fully engaged	1 %	
CpuUsageBG	float	1	%	Percent of available time bg thread took with a 1 sec avg	0.759 %	
CpuUsageFG	float	1	%	Percent of available time fg thread took with a 1 sec avg	0.6478 %	
DCDriversSoFar	int	1		Number of team drivers who have run a stint	1	
DCLapStatus	int	1		Status of driver change lap requirements	2	
DisplayUnits	int	1		Default units for the user interface 0 = english 1 = metric	1	
DriverMarker	bool	1		Driver activated flag	False	
EnergyBatteryToMGU_K	float	1	J	Electrical energy from battery to MGU-K	160206 J	

Nome	Tipo	Qt d	Unidade	Descrição	Exemplo (valor gravado na corrida)	Usada
KLap				per lap		
EnergyERSBattery	float	1	J	Engine ERS battery charge	4.6384e+06 J	
EnergyERSBatteryPct	float	1	%	Engine ERS battery charge as a percent	0.9544 %	
EnergyMGU_KLapDeployPct	float	1	%	Electrical energy available to MGU-K per lap as a percent	0 %	
Engine0_RPM	float	1	revs/min	Engine0Engine rpm	7621.11 revs/min	
Engine1_RPM	float	1	revs/min	Engine1Engine rpm	7621.11 revs/min	
EngineWarnings	bitfield	1	irsdk_EngineWarnings	Bitfield for warning lights	0 (0x0) = nenhum bit	
EnterExitReset	int	1		Indicate action the reset key will take 0 enter 1 exit 2 reset	2	
FastRepairAvailable	int	1		How many fast repairs left 255 is unlimited	0	
FastRepairUsed	int	1		How many fast repairs used so far	0	
FogLevel	float	1	%	Fog level at start/finish line	0 %	
FrameRate	float	1	fps	Average frames per second	90.5028 fps	
FrontTireSetsAvailable	int	1		How many front tire sets are remaining 255 is unlimited	1	
FrontTireSetsUsed	int	1		How many front tire sets used so far	1	
FuelLevel	float	1	l	Liters of fuel remaining	57.5142 l	sim
FuelLevelPct	float	1	%	Percent fuel remaining	0.6462 %	sim
FuelPress	float	1	bar	Engine fuel pressure	5.83 bar	
FuelUsePerHour	float	1	kg/h	Engine fuel used instantaneous	112.177 kg/h	sim
Gear	int	1		-1=reverse 0=neutral 1..n=current gear	4	sim

Nome	Tipo	Qt d	Unidade	Descrição	Exemplo (valor gravado na corrida)	Usada
GpuUsage	float	1	%	Percent of available time gpu took with a 1 sec avg	0.4401 %	
HFshockDefl	float	1	m	HF shock deflection	0.0465 m	
HFshockDefl_ST	float	6	m	HF shock deflection at 360 Hz	carros 0-3: [0.0458, 0.046, 0.0462, 0.0463] m	
HFshockVel	float	1	m/s	HF shock velocity	0.0346 m/s	
HFshockVel_ST	float	6	m/s	HF shock velocity at 360 Hz	carros 0-3: [0.0695, 0.0427, 0.0213, 0.0051] m/s	
HandbrakeRaw	float	1	%	Raw handbrake input 0=handbrake released to 1=max force	0 %	
IsDiskLoggingActive	bool	1		0=disk based telemetry file not being written 1=being written	True	
IsDiskLoggingEnabled	bool	1		0=disk based telemetry turned off 1=turned on	True	
IsGarageVisible	bool	1		1=Garage screen is visible	False	
IsInGarage	bool	1		1=Car in garage physics running	False	
IsOnTrack	bool	1		1=Car on track physics running with player in car	True	sim
IsOnTrackCar	bool	1		1=Car on track physics running	True	
IsReplayPlaying	bool	1		0=replay not playing 1=replay playing	False	
LFTiresAvailable	int	1		How many left front tires are remaining 255 is unlimited	1	
LFTiresUsed	int	1		How many left front tires used so far	1	
LFbrakeLinePress	float	1	bar	LF brake line pressure	0 bar	
LFcoldPressure	float	1	kPa	LF tire cold pressure as set in the garage	151.606 kPa	
LFodometer	float	1	m	LF distance tire traveled since being	100 m	

Nome	Tipo	Qtd	Unidade	Descrição	Exemplo (valor gravado na corrida)	Usada
				placed on car		
LFshockDefl	float	1	m	LF shock deflection	0.0158 m	
LFshockDefl_ST	float	6	m	LF shock deflection at 360 Hz	carros 0-3: [0.0156, 0.0157, 0.0157, 0.0157] m	
LFshockVel	float	1	m/s	LF shock velocity	0.0296 m/s	
LFshockVel_ST	float	6	m/s	LF shock velocity at 360 Hz	carros 0-3: [0.0299, 0.0197, 0.0105, 0.0084] m/s	
LFtempCL	float	1	C	LF tire left carcass temperature	78.3269 C	
LFtempCM	float	1	C	LF tire middle carcass temperature	83.6539 C	
LFtempCR	float	1	C	LF tire right carcass temperature	86.3208 C	
LFwearL	float	1	%	LF tire left percent tread remaining	0.9902 %	
LFwearM	float	1	%	LF tire middle percent tread remaining	0.9858 %	
LFwearR	float	1	%	LF tire right percent tread remaining	0.9862 %	sim
LRTiresAvailable	int	1		How many left rear tires are remaining 255 is unlimited	1	
LRTiresUsed	int	1		How many left rear tires used so far	1	
LRbrakeLinePress	float	1	bar	LR brake line pressure	0 bar	
LRcoldPressure	float	1	kPa	LR tire cold pressure as set in the garage	151.606 kPa	
LRodometer	float	1	m	LR distance tire traveled since being placed on car	100 m	
LRshockDefl	float	1	m	LR shock deflection	0.0302 m	
LRshockDefl_ST	float	6	m	LR shock deflection at 360 Hz	carros 0-3: [0.0299, 0.0299, 0.03, 0.03] m	
LRshockVel	float	1	m/s	LR shock velocity	0.0237 m/s	
LRshockVel_ST	float	6	m/s	LR shock velocity at 360 Hz	carros 0-3: [0.0051, 0.007, 0.0162, 0.0258] m/s	

Nome	Tipo	Qt d	Unidade	Descrição	Exemplo (valor gravado na corrida)	Usada
LRtempCL	float	1	C	LR tire left carcass temperature	81.8933 C	
LRtempCM	float	1	C	LR tire middle carcass temperature	86.2442 C	
LRtempCR	float	1	C	LR tire right carcass temperature	87.7234 C	
LRwearL	float	1	%	LR tire left percent tread remaining	0.983 %	
LRwearM	float	1	%	LR tire middle percent tread remaining	0.9787 %	
LRwearR	float	1	%	LR tire right percent tread remaining	0.9793 %	
Lap	int	1		Laps started count	1	sim
LapBestLap	int	1		Players best lap number	0	
LapBestLapTime	float	1	s	Players best lap time	0 s	
LapBestNLapLap	int	1		Player last lap in best N average lap time	0	
LapBestNLapTime	float	1	s	Player best N average lap time	0 s	
LapCompleted	int	1		Laps completed count	0	
LapCurrentLapTime	float	1	s	Estimate of players current lap time as shown in F3 box	7.4626 s	
LapDeltaToBestLap	float	1	s	Delta time for best lap	1.7787 s	
LapDeltaToBestLap_DD	float	1	s/s	Rate of change of delta time for best lap	0.3194 s/s	
LapDeltaToBestLap_OK	bool	1		Delta time for best lap is valid	True	
LapDeltaToOptimalLap	float	1	s	Delta time for optimal lap	1.7805 s	
LapDeltaToOptimalLap_DD	float	1	s/s	Rate of change of delta time for optimal lap	0.3195 s/s	
LapDeltaToOptimalLap_OK	bool	1		Delta time for optimal lap is valid	True	
LapDeltaToSessionBestLap	float	1	s	Delta time for session best lap	0 s	

Nome	Tipo	Qt d	Unidade	Descrição	Exemplo (valor gravado na corrida)	Usada
LapDeltaToSessionBestLap_DD	float	1	s/s	Rate of change of delta time for session best lap	0 s/s	
LapDeltaToSessionBestLap_OK	bool	1		Delta time for session best lap is valid	False	
LapDeltaToSessionLastLap	float	1	s	Delta time for session last lap	0 s	
LapDeltaToSessionLastLap_DD	float	1	s/s	Rate of change of delta time for session last lap	0 s/s	
LapDeltaToSessionLastLap_OK	bool	1		Delta time for session last lap is valid	False	
LapDeltaToSessionOptimalLap	float	1	s	Delta time for session optimal lap	0 s	
LapDeltaToSessionOptimalLap_DD	float	1	s/s	Rate of change of delta time for session optimal lap	0 s/s	
LapDeltaToSessionOptimalLap_OK	bool	1		Delta time for session optimal lap is valid	False	
LapDist	float	1	m	Meters traveled from S/F this lap	145.209 m	
LapDistPct	float	1	%	Percentage distance around lap	0.0358 %	sim
LapLasNLapSeq	int	1		Player num consecutive clean laps completed for N average	0	
LapLastLapTime	float	1	s	Players last lap time	0 s	sim
LapLastNLapTime	float	1	s	Player last N average lap time	0 s	
LatAccel	float	1	m/s^2	Lateral acceleration (including gravity)	3.4043 m/s^2	
LatAccel_ST	float	6	m/s^2	Lateral acceleration (including gravity) at 360 Hz	carros 0-3: [3.624, 3.5692, 3.5567, 3.542] m/s^2	

Nome	Tipo	Qt d	Unidade	Descrição	Exemplo (valor gravado na corrida)	Usa da
LeftTireSetsAvailable	int	1		How many left tire sets are remaining 255 is unlimited	1	
LeftTireSetsUsed	int	1		How many left tire sets used so far	1	
LoadNumTextures	bool	1		True if the car_num texture will be loaded	False	
LongAccel	float	1	m/s^2	Longitudinal acceleration (including gravity)	5.6845 m/s^2	
LongAccel_ST	float	6	m/s^2	Longitudinal acceleration (including gravity) at 360 Hz	carros 0-3: [5.5564, 5.5619, 5.582, 5.618] m/s^2	
ManifoldPress	float	1	bar	Engine manifold pressure	0.9816 bar	
ManualBoost	bool	1		Hybrid manual boost state	False	
ManualNoBoost	bool	1		Hybrid manual no boost state	False	
MemPageFaultSec	float	1		Memory page faults per second	0	
MemSoftPageFaultSec	float	1		Memory soft page faults per second	0	
OilLevel	float	1	l	Engine oil level	12.68 l	
OilPress	float	1	bar	Engine oil pressure	5.24 bar	
OilTemp	float	1	C	Engine oil temperature	84.6924 C	
OkToReloadTextures	bool	1		True if it is ok to reload car textures at this time	True	
OnPitRoad	bool	1		Is the player car on pit road between the cones	False	sim
P2P_Count	int	1		Push2Pass count of usage (or remaining in Race) on your car	0	sim
P2P_Status	bool	1		Push2Pass active or not on your car	False	
PaceMode	int	1	irsdk_PaceMode	Are we pacing or not	1 = DoubleFileStart	

Nome	Tipo	Qt d	Unidade	Descrição	Exemplo (valor gravado na corrida)	Usada
PitOptRepairLeft	float	1	s	Time left for optional repairs if repairs are active	0 s	
PitRepairLeft	float	1	s	Time left for mandatory pit repairs if repairs are active	0 s	
PitSvFlags	bitfield	1	irsdk_PitSvFlags	Bitfield of pit service checkboxes	16 (0x10) = fuel	sim
PitSvFuel	float	1	l or kWh	Pit service fuel add amount	34 l or kWh	sim
PitSvLFP	float	1	kPa	Pit service left front tire pressure	151.606 kPa	
PitSvLRP	float	1	kPa	Pit service left rear tire pressure	151.606 kPa	
PitSvRFP	float	1	kPa	Pit service right front tire pressure	151.606 kPa	
PitSvRRP	float	1	kPa	Pit service right rear tire pressure	151.606 kPa	
PitSvTireCompound	int	1		Pit service pending tire compound	0	
Pitch	float	1	rad	Pitch orientation	0.0125 rad	
PitchRate	float	1	rad/s	Pitch rate	-0.0567 rad/s	
PitchRate_ST	float	6	rad/s	Pitch rate at 360 Hz	carros 0-3: [-0.0419, -0.0469, -0.0509, -0.0536] rad/s	
PitsOpen	bool	1		True if pit stop is allowed for the current player	True	
PitstopActive	bool	1		Is the player getting pit stop service	False	
PlayerCarClass	int	1		Player car class id	4029	
PlayerCarClassPosition	int	1		Players class position in race	5	
PlayerCarDriverIncidentCount	int	1		Teams current drivers incident count for this session	0	
PlayerCarDryTireSetLimit	int	1		Players dry tire set limit	0	

Nome	Tipo	Qt d	Unidade	Descrição	Exemplo (valor gravado na corrida)	Usa da
PlayerCarIdx	int	1		Players carIdx	0	sim
PlayerCarInPitStall	bool	1		Players car is properly in their pitstall	False	
PlayerCarMyIncidentCount	int	1		Players own incident count for this session	0	sim
PlayerCarPitSvStatus	int	1	irsdk_PitSvStatus	Players car pit service status bits	0 = None	
PlayerCarPosition	int	1		Players position in race	5	
PlayerCarPowerAdjust	float	1	%	Players power adjust	0 %	
PlayerCarSLBlinkRPM	float	1	revs/min	Shift light blink rpm	8725 revs/min	
PlayerCarSLFirstRPM	float	1	revs/min	Shift light first light rpm	8300 revs/min	
PlayerCarSLLastRPM	float	1	revs/min	Shift light last light rpm	8650 revs/min	
PlayerCarSLShiftRPM	float	1	revs/min	Shift light shift rpm	8550 revs/min	
PlayerCarTeamIncidentCount	int	1		Players team incident count for this session	0	
PlayerCarTowTime	float	1	s	Players car is being towed if time is greater than zero	0 s	sim
PlayerCarWeightPenalty	float	1	kg	Players weight penalty	0 kg	
PlayerFastRepairsUsed	int	1		Players car number of fast repairs used	0	
PlayerIncidents	int	1	irsdk_IncidentFlags	Log incidents that the player recieved	0	
PlayerTireCompound	int	1		Players car current tire compound	0	
PlayerTrackSurface	int	1	irsdk_TrkLoc	Players car track surface type	3 = OnTrack	sim
PlayerTrackSurfaceMaterial	int	1	irsdk_TrkSurf	Players car track surface material type	1	

Nome	Tipo	Qt d	Unidade	Descrição	Exemplo (valor gravado na corrida)	Usada
PowerMGU_H	float	1	W	Engine MGU-H mechanical power	-0 W	
PowerMGU_K	float	1	W	Engine MGU-K mechanical power	40617 W	
Precipitation	float	1	%	Precipitation at start/finish line	0 %	sim
PushToPass	bool	1		Push to pass button state	False	
PushToTalk	bool	1		Push to talk button state	False	
RFTiresAvailable	int	1		How many right front tires are remaining 255 is unlimited	1	
RFTiresUsed	int	1		How many right front tires used so far	1	
RFbrakeLinePress	float	1	bar	RF brake line pressure	0 bar	
RFcoldPressure	float	1	kPa	RF tire cold pressure as set in the garage	151.606 kPa	
RFodometer	float	1	m	RF distance tire traveled since being placed on car	100 m	
RFshockDefl	float	1	m	RF shock deflection	0.0177 m	
RFshockDefl_ST	float	6	m	RF shock deflection at 360 Hz	carros 0-3: [0.0174, 0.0175, 0.0175, 0.0176] m	
RFshockVel	float	1	m/s	RF shock velocity	-0.0047 m/s	
RFshockVel_ST	float	6	m/s	RF shock velocity at 360 Hz	carros 0-3: [0.0201, 0.011, 0.0048, -0.0047] m/s	
RFtempCL	float	1	C	RF tire left carcass temperature	78.2775 C	
RFtempCM	float	1	C	RF tire middle carcass temperature	75.7167 C	
RFtempCR	float	1	C	RF tire right carcass temperature	66.3126 C	
RFwearL	float	1	%	RF tire left percent tread remaining	0.9887 %	
RFwearM	float	1	%	RF tire middle percent tread remaining	0.9882 %	
RFwearR	float	1	%	RF tire right percent tread remaining	0.9955 %	

Nome	Tipo	Qt d	Unidade	Descrição	Exemplo (valor gravado na corrida)	Usada
RPM	float	1	revs/min	Engine rpm	7621.1 revs/min	sim
RRTiresAvailable	int	1		How many right rear tires are remaining 255 is unlimited	1	
RRTiresUsed	int	1		How many right rear tires used so far	1	
RRbrakeLinePress	float	1	bar	RR brake line pressure	0 bar	
RRcoldPressure	float	1	kPa	RR tire cold pressure as set in the garage	151.606 kPa	
RRodometer	float	1	m	RR distance tire traveled since being placed on car	100 m	
RRshockDefl	float	1	m	RR shock deflection	0.0318 m	
RRshockDefl_ST	float	6	m	RR shock deflection at 360 Hz	carros 0-3: [0.0318, 0.0318, 0.0317, 0.0317] m	
RRshockVel	float	1	m/s	RR shock velocity	0.0192 m/s	
RRshockVel_ST	float	6	m/s	RR shock velocity at 360 Hz	carros 0-3: [-0.0128, -0.0056, 0.0034, 0.0108] m/s	
RRtempCL	float	1	C	RR tire left carcass temperature	82.3492 C	
RRtempCM	float	1	C	RR tire middle carcass temperature	80.2213 C	
RRtempCR	float	1	C	RR tire right carcass temperature	70.8238 C	
RRwearL	float	1	%	RR tire left percent tread remaining	0.9818 %	
RRwearM	float	1	%	RR tire middle percent tread remaining	0.9814 %	
RRwearR	float	1	%	RR tire right percent tread remaining	0.9929 %	
RaceLaps	int	1		Laps completed in race	1	
RadioTransmitCarIdx	int	1		The car index of the current person speaking on the radio	-1	
RadioTransmitFrequencyIdx	int	1		The frequency index of the current person speaking on the radio	1	

Nome	Tipo	Qt d	Unidade	Descrição	Exemplo (valor gravado na corrida)	Usada
RadioTransmitRadioldx	int	1		The radio index of the current person speaking on the radio	0	
RearTireSetsAvailable	int	1		How many rear tire sets are remaining 255 is unlimited	1	
RearTireSetsUsed	int	1		How many rear tire sets used so far	1	
RelativeHumidity	float	1	%	Relative Humidity at start/finish line	0.7319 %	sim
ReplayFrameNum	int	1		Integer replay frame number (60 per second)	0	
ReplayFrameNumEnd	int	1		Integer replay frame number from end of tape	28086	
ReplayPlaySlowMotion	bool	1		0=not slow motion 1=replay is in slow motion	False	
ReplayPlaySpeed	int	1		Replay playback speed	1	
ReplaySessionNum	int	1		Replay session number	-1	
ReplaySessionTime	double	1	s	Seconds since replay session start	0 s	
RightTireSetsAvailable	int	1		How many right tire sets are remaining 255 is unlimited	1	
RightTireSetsUsed	int	1		How many right tire sets used so far	1	
Roll	float	1	rad	Roll orientation	0.0069 rad	
RollRate	float	1	rad/s	Roll rate	0.0124 rad/s	
RollRate_ST	float	6	rad/s	Roll rate at 360 Hz	carros 0-3: [-0.0017, 0.0001, 0.0018, 0.0028] rad/s	
SessionFlags	bitfield	1	irsdk_Flags	Session flags	268697636 (0x10040024) = green + blue + servicable + startHidden	sim

Nome	Tipo	Qt d	Unidade	Descrição	Exemplo (valor gravado na corrida)	Usada
SessionJokerLapsRemain	int	1		Joker laps remaining to be taken	0	
SessionLapsRemain	int	1		Old laps left till session ends use SessionLapsRemainEx	32767	
SessionLapsRemainEx	int	1		New improved laps left till session ends	32767	
SessionLapsTotal	int	1		Total number of laps in session	32767	
SessionNum	int	1		Session number	2	sim
SessionOnJokerLap	bool	1		Player is currently completing a joker lap	False	
SessionState	int	1	irsdk_SessionState	Session state	4 = Racing	sim
SessionTick	int	1		Current update number	40078	
SessionTime	double	1	s	Seconds since session start	87.2833 s	sim
SessionTimeOfDay	float	1	s	Time of day in seconds	63867 s	
SessionTimeRemain	double	1	s	Seconds left till session ends	2692.52 s	sim
SessionTimeTotal	double	1	s	Total number of seconds in session	2700 s	
SessionUniqueID	int	1		Session ID	3	sim
ShiftGrindRPM	float	1	RPM	RPM of shifter grinding noise	0 RPM	
ShiftIndicatorPct	float	1	%	DEPRECATED use DriverCarSLBlinkRPM instead	0 %	
ShiftPowerPct	float	1	%	Friction torque applied to gears when shifting or grinding	0 %	
Shifter	int	1		Log inputs from the players shifter control	0	

Nome	Tipo	Qt d	Unidade	Descrição	Exemplo (valor gravado na corrida)	Usa da
Skies	int	1		Skies (0=clear/1=p cloudy/2=m cloudy/3=overcast)	1	
SolarAltitude	float	1	rad	Sun angle above horizon in radians	0.3483 rad	
SolarAzimuth	float	1	rad	Sun angle clockwise from north in radians	4.4263 rad	
Speed	float	1	m/s	GPS vehicle speed	52.8334 m/s	sim
SteeringFFBEnabled	bool	1		Force feedback is enabled	True	
SteeringWheelAngle	float	1	rad	Steering wheel angle	0.0639 rad	sim
SteeringWheelAngleMax	float	1	rad	Steering wheel max angle	9.4255 rad	
SteeringWheelLimiter	float	1	%	Force feedback limiter strength limits impacts and oscillation	0 %	
SteeringWheelMaxForceNm	float	1	N*m	Value of strength or max force slider in Nm for FFB	33.2546 N*m	
SteeringWheelPctDamp	float	1	%	Force feedback % max damping	0 %	
SteeringWheelPctIntensity	float	1	%	Force feedback % max intensity	1 %	
SteeringWheelPctSmoothing	float	1	%	Force feedback % max smoothing	0 %	
SteeringWheelPctTorque	float	1	%	Force feedback % max torque on steering shaft unsigned	0.1174 %	
SteeringWheelPctTorqueSign	float	1	%	Force feedback % max torque on steering shaft signed	-0.1174 %	
SteeringWheelPctTorqueSignStops	float	1	%	Force feedback % max torque on steering shaft signed stops	-0.1174 %	

Nome	Tipo	Qt d	Unidade	Descrição	Exemplo (valor gravado na corrida)	Usa da
SteeringWheelPeakForceNm	float	1	N*m	Peak torque mapping to direct input units for FFB	-1 N*m	
SteeringWheelTorque	float	1	N*m	Output torque on steering shaft	-3.9047 N*m	
SteeringWheelTorque_ST	float	6	N*m	Output torque on steering shaft at 360 Hz	carros 0-3: [-3.8967, -3.8869, -3.9816, -4.015] N*m	
SteeringWheelUseLinear	bool	1		True if steering wheel force is using linear mode	True	
Throttle	float	1	%	0=off throttle to 1=full throttle	1 %	sim
ThrottleRaw	float	1	%	Raw throttle input 0=off throttle to 1=full throttle	1 %	
TireLF_RumblePitch	float	1	Hz	Players LF Tire Sound rumblestrip pitch	0 Hz	
TireLR_RumblePitch	float	1	Hz	Players LR Tire Sound rumblestrip pitch	0 Hz	
TireRF_RumblePitch	float	1	Hz	Players RF Tire Sound rumblestrip pitch	0 Hz	
TireRR_RumblePitch	float	1	Hz	Players RR Tire Sound rumblestrip pitch	0 Hz	
TireSetsAvailable	int	1		How many tire sets are remaining 255 is unlimited	1	
TireSetsUsed	int	1		How many tire sets used so far	1	
TorqueMGU_K	float	1	Nm	Engine MGU-K mechanical torque	50.8934 Nm	
TrackTemp	float	1	C	Deprecated set to TrackTempCrew	26.6667 C	sim
TrackTempCrew	float	1	C	Temperature of track measured by crew around track	26.6667 C	
TrackWetness	int	1	irsdk_TrackWetness	How wet is the average track surface	1 = Dry	sim
VelocityX	float	1	m/s	X velocity	52.8333 m/s	
VelocityX_ST	float	6	m/s at 360 Hz	X velocity	carros 0-3: [52.7536, 52.7693, 52.7852,	

Nome	Tipo	Qt d	Unidade	Descrição	Exemplo (valor gravado na corrida)	Usada
					52.8011] m/s at 360 Hz	
VelocityY	float	1	m/s	Y velocity	-0.0797 m/s	
VelocityY_ST	float	6	m/s at 360 Hz	Y velocity	carros 0-3: [-0.0708, -0.0723, -0.0739, -0.0756] m/s at 360 Hz	
VelocityZ	float	1	m/s	Z velocity	0.0488 m/s	
VelocityZ_ST	float	6	m/s at 360 Hz	Z velocity	carros 0-3: [0.0676, 0.0648, 0.0611, 0.057] m/s at 360 Hz	
VertAccel	float	1	m/s^2	Vertical acceleration (including gravity)	11.3071 m/s^2	
VertAccel_ST	float	6	m/s^2	Vertical acceleration (including gravity) at 360 Hz	carros 0-3: [11.2555, 11.1391, 11.0165, 11.0499] m/s^2	
VidCapActive	bool	1		True if video currently being captured	False	
VidCapEnabled	bool	1		True if video capture system is enabled	False	
Voltage	float	1	V	Engine voltage	13.4 V	
WaterLevel	float	1	l	Engine coolant level	7 l	
WaterTemp	float	1	C	Engine coolant temp	89.3046 C	
WeatherDeclaredWet	bool	1		The steward says rain tires can be used	False	sim
WindDir	float	1	rad	Wind direction at start/finish line	2.8629 rad	sim
WindVel	float	1	m/s	Wind velocity at start/finish line	4.4698 m/s	sim
Yaw	float	1	rad	Yaw orientation	-1.5528 rad	
YawNorth	float	1	rad	Yaw orientation relative to north	2.1131 rad	
YawRate	float	1	rad/s	Yaw rate	0.0781 rad/s	
YawRate_ST	float	6	rad/s	Yaw rate at 360 Hz	carros 0-3: [0.0758, 0.0766, 0.0772, 0.0777] rad/s	
dcABS	float	1		In car abs adjustment	0	
dcAntiRollFront	float	1		In car front anti roll bar adjustment	1	

Nome	Tipo	Qt d	Unidade	Descrição	Exemplo (valor gravado na corrida)	Usa da
dcAntiRollRear	float	1		In car rear anti roll bar adjustment	1	
dcBrakeBias	float	1		In car brake bias adjustment	47	sim
dcBrakeMisc	float	1		In car brake misc adjustment	0	
dcHeadlightFlash	bool	1		In car headlight flash control active	False	
dcLowFuelAccept	bool	1		In car low fuel accept	False	
dcPitSpeedLimiterToggle	bool	1		Track if pit speed limiter system is enabled	False	
dcStarter	bool	1		In car trigger car starter	False	
dcToggleWindshieldWipers	bool	1		In car turn wipers on or off	False	
dcTractionControl	float	1		In car traction control adjustment	4	
dcTractionControl2	float	1		In car traction control 2 adjustment	4	
dcTractionControlToggle	bool	1		In car traction control active	False	
dcTriggerWindshieldWipers	bool	1		In car momentarily turn on wipers	False	
dpFastRepair	float	1		Pitstop fast repair set	0	
dpFuelAddKg	float	1	kg	Pitstop fuel add amount	34 kg	
dpFuelAutoFillActive	float	1		Pitstop auto fill fuel next stop flag	1	
dpFuelAutoFillEnabled	float	1		Pitstop auto fill fuel system enabled	1	
dpFuelFill	float	1		Pitstop fuel fill flag	1	
dpLFTireChange	float	1		Pitstop If tire change request	0	
dpLFTireColdPress	float	1	Pa	Pitstop If tire cold pressure adjustment	151.606 Pa	
dpLRTireChange	float	1		Pitstop Ir tire change request	0	

Nome	Tipo	Qt d	Unidade	Descrição	Exemplo (valor gravado na corrida)	Usa da
dpLRTireColdPress	float	1	Pa	Pitstop lr tire cold pressure adjustment	151.606 Pa	
dpRFTireChange	float	1		Pitstop rf tire change request	0	
dpRFTireColdPress	float	1	Pa	Pitstop rf cold tire pressure adjustment	151.606 Pa	
dpRRTireChange	float	1		Pitstop rr tire change request	0	
dpRRTireColdPress	float	1	Pa	Pitstop rr cold tire pressure adjustment	151.606 Pa	
dpWindshieldTearoff	float	1		Pitstop windshield tearoff	0	

SessionInfo (seções)

Seção	Chaves
WeekendInfo	62
SessionInfo	2
QualifyResultsInfo	1
CameraInfo	1
RadiolInfo	2
DriverInfo	35
SplitTimeInfo	1
CarSetup	4